AN ALGEBRA OVER A FIELD IS ETALE EXACTLY WHEN IT IS A FINITE PRODUCT OF FINITE SEPARABLE EXTENSIONS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Etaleness is stable under finite products. Used at the start of the counted proof (lines 39618-39621). Reduces the statement to the individual factors of a finite product.
- Smoothness at the generic point forces separability. Invoked at the end of the counted proof (lines 39646-39648). Shows the field extension is separable once the algebra is smooth.

Source reference. Original lemma 00TT: sources/stacks-algebra/pinned-original-source.tex, source lines 38826–38826. Navigation pointer for \ref{lemma-characterize-smooth-over-field} used inside the counted span (line 39636); the argument that consumes it is retained in the span.

Source reference. Original lemma 00U2: sources/stacks-algebra/pinned-original-source.tex, source lines 39507–39507. Navigation pointer for \ref{lemma-etale} used inside the counted span (line 39621); the argument that consumes it is retained in the span.

Date: stacks-project:algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 07ND: sources/stacks-algebra/pinned-original-source.tex, source lines 39108–39108. Navigation pointer for \ref{lemma-smooth-at-generic-point} used inside the counted span (line 39646); the argument that consumes it is retained in the span.

Source reference. Original lemma 00KJ: sources/stacks-algebra/pinned-original-source.tex, source lines 14632–14632. Navigation pointer for \ref{proposition-dimension-zero-ring} used inside the counted span (line 39639); the argument that consumes it is retained in the span.

Source reference. Original lemma 00FV: sources/stacks-algebra/pinned-original-source.tex, source lines 6741–6741. Navigation pointer for \ref{theorem-nullstellensatz} used inside the counted span (line 39644); the argument that consumes it is retained in the span.

Lemma 0.1. *Let k be a field. A ring map $k \to S$ is étale if and only if S is isomorphic as a k-algebra to a finite product of finite separable extensions of k.*

Proof. We are going to use without further mention: if $S = S_1 \times \ldots \times S_n$ is a finite product of k-algebras, then S is étale over k if and only if each S_i is étale over k. See Lemma 00U2 part (11).

If k'/k is a finite separable field extension then we can write $k' = k(\alpha) \cong k[x]/(f)$. Here f is the minimal polynomial of the element α. Since k' is separable over k we have $\gcd(f, f') = 1$. This implies that $\mathrm{d} : k' \cdot f \to k' \cdot \mathrm{d}x$ is an isomorphism. Hence $k \to k'$ is étale. Thus if S is a finite product of finite separable extensions of k, then S is étale over k.

Conversely, suppose that $k \to S$ is étale. Then S is smooth over k and $\Omega_{S/k} = 0$. By Lemma 00TT we see that $\dim_{\mathfrak{m}} \mathrm{Spec}(S) = 0$ for every maximal ideal $\mathfrak{m}$ of S. Thus $\dim(S) = 0$. By Proposition 00KJ we find that S is a finite product of Artinian local rings. By the already used Lemma 00TT these local rings are fields. Hence we may assume $S = k'$ is a field. By the Hilbert Nullstellensatz (Theorem 00FV) we see that the extension k'/k is finite. The smoothness of $k \to k'$ implies by Lemma 07ND that k'/k is a separable extension and the proof is complete. □